Guide for participants

Structured Methodology

Purpose

1. This methodology provides **a structured pathway** for the teams to go from challenge → solution.
2. Use the **questions and templates** in the way that best supports your team's thinking. You do not need to complete every activity or answer every question.
3. The methodology is grounded in a **human centred approach**, helping teams understand real user needs, their challenges, and decision-making process before designing solutions to **create meaningful value for users**.
4. The aim is to co-create solutions that are **desirable for real users**, valuable to stakeholders, and enabled by data.

Guided Methodology

CATAPULT Connected Places

- Define Your Challenge & Explore Data
- Stakeholder Mapping
- Define Your User Persona
- Use Case & User Journey
- Prototyping & User Validation
- Roadmap & Next Steps
- Slide Template

Define your Challenge

Objective: Align the team around a meaningful and achievable challenge.

Questions:

- Which theme resonates most with your team?
- What specific challenge will you focus on today?
- Why is this challenge important?
- Who experiences this challenge?
- What happens if this challenge is not addressed?
- What opportunities could be unlocked if it were solved?

Selected Theme:

Enter Theme Here

Challenge Statement:

How might we ____________ so that ____________ can ____________?

Explore Data

Objective: Understand what data is available, who it is valuable to, and how it could help address your chosen challenge.

Questions:
- What datasets are available that relate to your chosen challenge?
- Which dataset or combination of datasets offers the opportunity to solve your challenge?
- What information, insights, or trends can these datasets provide?
- What decisions could be informed using this data? What decisions depend on this data?
- Who makes decisions and could benefit from this data?
- How could this data be transformed into actionable outputs or services for users?
- What initial solution ideas do you have?

Data Exploration

Available Data	Key Insights	Decision Supported
Text	Text	Text
Text	Text	Text
Text	Text	Text
Text	Text	Text
Text	Text	Text
Text	Text	Text

Stakeholder Mapping

Objective: Understand who is affected by the challenge and or datasets.

Questions:

- Who is directly affected?
- Who is indirectly affected?
- Who makes decisions?
- Who would benefit?
- Who could be negatively impacted?

Stakeholder Map

Low impact / indirect involvement

Medium impact / indirect involvement

High impact / direct user

E.g. Local Councils

Define Your User Persona

Objective: Build empathy and focus the solution on a real user's needs.

Questions:

- About the User: Who is your user? What is their role and organisation they work at? What are their key responsibilities? What goals are they trying to achieve?
- Current Ways of Working: What tools, systems, or resources do they currently use? What information or data do they rely on? What outputs or decisions are they responsible for?
- Challenges: What frustrations, barriers, or pain points do they face? What data do they need? What information is difficult to access or understand?
- Future Value: How could your solution help them? What benefits would they experience? What would success look like from their perspective?

Space for photograph

User Persona

Name here

"As a …I need to … and will help me do … better"

Role:
Type here

Organisation:
Type here

Tools & resources:
- Type here

Responsibilities & Goals:
- Type here

Challenges:
- Type here

Data/Outputs Required:
- Type here

Potential Benefits
- Type here

Design thinking is a **human centered approach to innovation** that draws from the designer's toolkit to integrate the **needs of people**, the **possibilities of technology**, and the requirements for business success.

Tim Brown,
Former IDEO CEO

Use Case and User Journey

Objective: Understand how work happens today and how your solution improves it.

Part A: Current State

Questions
- What is the user trying to achieve?
- What steps do they currently follow?
- What data or information do they use at each step?
- Where do delays, frustrations, or inefficiencies occur?
- What decisions must they make along the way?

Current User Journey (Without Solution)

User Steps / Actions	Add step or action here	Add step or action here	Add step or action here	Add step or action here	Add step or action here	Add step or action here	Add step or action here
Data input / Tools used	Add here	Add here	Add here	Add here	Add here	Add here	Add here
Challenges/ Pain Points	Add here	Add here	Add here	Add here	Add here	Add here	Add here
Decisions Made	Add here	Add here	Add here	Add here	Add here	Add here	Add here

Use Case and User Journey

Objective: Understand how work happens today and how your solution improves it.

Part B: Future State

Questions

- How does your solution fit into the user's workflow?
- What are the key steps the user takes with your solution?
- What data goes in?
- What outputs come out?
- What decisions does the solution support?
- How does the user benefit from the solution and outputs?

Future User Journey (With Solution)

User Steps / Actions	Add step or action here	Add step or action here	Add step or action here	Add step or action here	Add step or action here	Add step or action here	Add step or action here
Data input	Add here	Add here	Add here	Add here	Add here	Add here	Add here
Outputs	Add here	Add here	Add here	Add here	Add here	Add here	Add here
Benefits	Add here	Add here	Add here	Add here	Add here	Add here	Add here

Prototype & Validation

Objective: Bring your idea to life and gather feedback quickly (if possible).

Questions

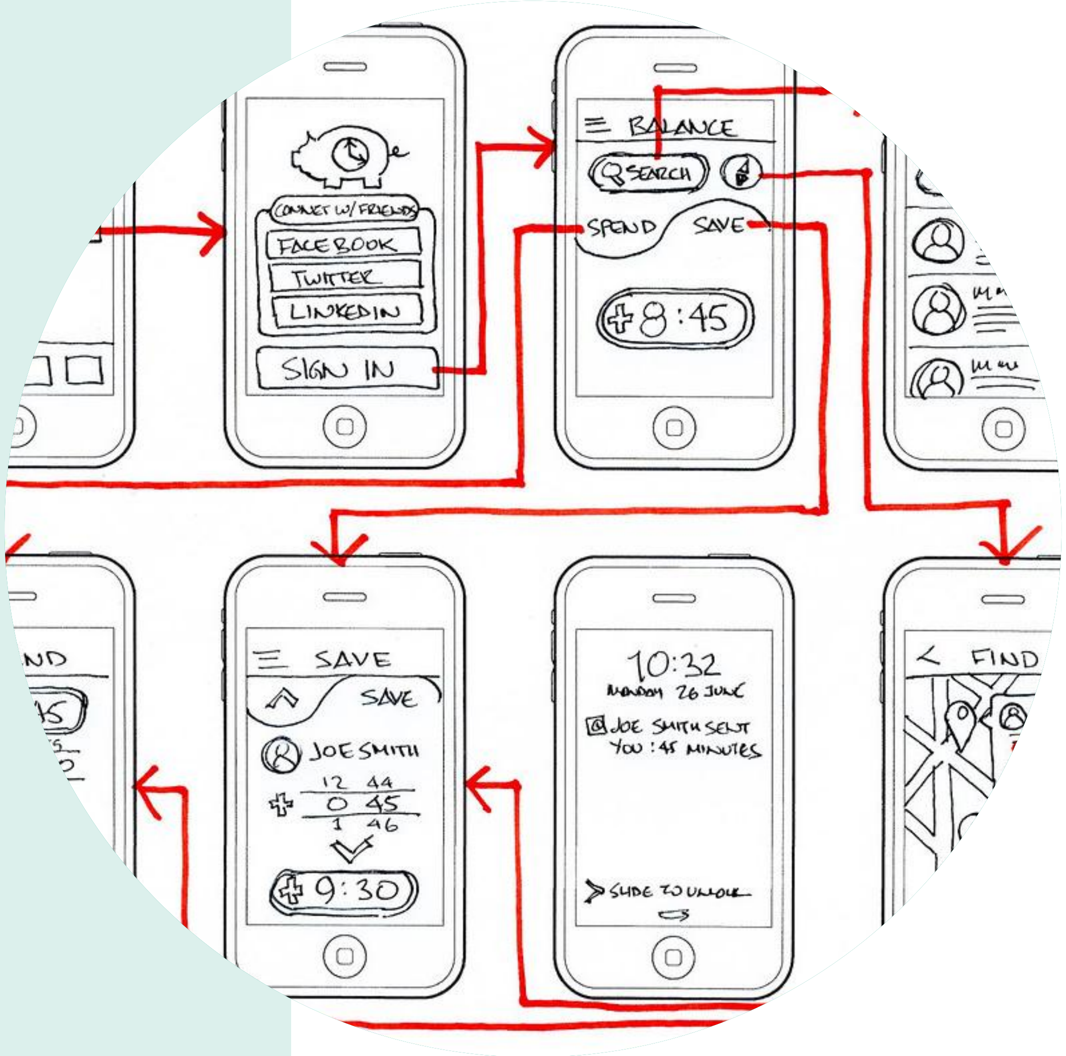
- What is the core capability of your solution?
- What problem does each feature solve?
- What data is needed for the solution to work?
- What outputs are generated?

User Validation (optional)

- Test the concept with the user or another participant.
- What worked well? What was confusing? What was missing?
- What improvements would you make with more time?

Build your prototype

- Create a wireframe or storyboard.
- Develop your prototype using Snowflake and available tools.



Roadmap & Next Steps

Objective: Show how the idea could move beyond the hackathon.

Questions

- What is the next step after today?
- What assumptions still need to be tested?
- What data is still required?
- What technical development is needed?
- Who needs to be involved next?
- What risks or barriers need addressing?
- What would a roadmap look like?
- How would you measure success?

Demonstrations, Presentations & Judging

Each team will be expected to present their prototypes or ideas developed in the Build sessions. The **presentations will last a maximum 5 minutes** per team. It is recommended that each team use the template provided to them. Each presentation will be evaluated by a judging panel. The presentations will be assessed by the following criteria:

Criteria

The criteria will be used by our judges to assess and select the winning team from the day. The winning idea will be considered by the Met Office and potentially advanced into alpha stage production. **Each criteria will be scored on a rating of 1-5**, with the following weightings:

- Technology & Innovation (50%)
- Impact & Value (30%)
- Feasibility (20%)

Criterion Overview

Technology & Innovation (50%)

- The originality, creativity and improvement to existing Met Office services. The strength of the approach, and the potential for increasing data use and access.

Impact & Value (30%)

- The extent to which the solution addresses the challenge or use case identified, and what benefits or value it creates for users, organisations, or society.

Feasibility (20%)

- The practicality of implementing the solution into existing systems, including the operational management and maintenance of the service improvement.

Presentation Slide Template

- 5 slides per team
- Minimum 2 slides showcasing solution/idea.

Selected Theme:

Climate Resilience and Public Services

Challenge Statement:

How might we **use weather & climate data** so that **future housing** can be **resilient to extreme weather events**?

Sarah Flakes

As a Senior Urban Planner, I need access to trusted data integrated with housing and planning information so that I can identify resilient development locations, reduce future climate risks, and make more evidence-based planning decisions.

Role:
Senior Urban Planning

Organisation:
Combined Authority X

Tools & resources:

- GIS and spatial planning platforms.
- UK Climate Projections (UKCP). Environment Agency flood risk maps.
- Local authority planning and housing data.
- Building and sustainability standards guidance.

Responsibilities & Goals:

- Develop regional spatial plans and housing strategies.
- Ensure new housing developments are sustainable and climate resilient.
- Coordinate with local authorities, developers, utilities, and environmental agencies.
- Support delivery of government housing targets while reducing climate related risks..

Challenges:

- Climate risks are often assessed inconsistently across local authorities.
- Climate and weather datasets are fragmented across multiple organisations.
- Balancing housing demand, affordability, environmental constraints, and resilience requirements.

Data/Outputs Required:

- Climate change scenario modelling for 2030, 2050, and 2080 at site & neighbourhood level.
- Surface water drainage and groundwater risk assessments.
- Housing site suitability and resilience scores.
- Visual dashboards showing climate risks by location. Impact assessments for different planning and design interventions.

Potential Benefits

- Better informed site selection for future housing.
- Reduced exposure of residents to flooding, overheating, and severe weather events.
- Lower long term retrofit and maintenance costs.
- Stronger evidence base for planning decisions.
- More resilient communities and infrastructure.
- Improved collaboration between planners, developers, utilities, and environmental stakeholders.

Thank you

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